

The empirical recurrence for OEIS A237144: recovering a conjecture that is not written down

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Abstract

OEIS A237144 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 216$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237144 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1472, 17616, 197724, 2463178, 30395540, 386318536, 4907807434, 63050866982, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:53:12, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 216$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 216$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 432$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	39	c_{19}	1554555314427677603	c_{37}	382888187578277599439919469	c_{55}
c_2	−160	c_{20}	−4037104924465079240	c_{38}	88741065470822620977868868	c_{56}
c_3	−12044	c_{21}	−23439383477887989510	c_{39}	−1330364967974923022020690482	c_{57}
c_4	145902	c_{22}	94468254131551344798	c_{40}	263612103359188023210204998	c_{58}
c_5	1301357	c_{23}	245986355248174699197	c_{41}	3632822012287157342790093062	c_{59}
c_6	−28998486	c_{24}	−1538068456145135501031	c_{42}	−1959772874708864128893661132	c_{60}
c_7	−23143247	c_{25}	−1509409575078360285981	c_{43}	−7800203620907892204716451725	c_{61}
c_8	2983103077	c_{26}	18585206013649610854908	c_{44}	6553800746731419550738384340	c_{62}
c_9	−8372689257	c_{27}	−969679345185840587997	c_{45}	13090078467385195437896000593	c_{63}
c_{10}	−186522916060	c_{28}	−171019089996424356270276	c_{46}	−14918802129637674735916961289	c_{64}
c_{11}	1021940846958	c_{29}	141770311292210122719180	c_{47}	−16929578769554397337783200256	c_{65}
c_{12}	7344299435320	c_{30}	1208287224153788793868230	c_{48}	25163812912874646295081404754	c_{66}
c_{13}	−63749321988945	c_{31}	−1833180013932432304366755	c_{49}	16411352874801018496786413612	c_{67}
c_{14}	−166330077893225	c_{32}	−6515970743144530671621596	c_{50}	−32344378109509927413894038376	c_{68}
c_{15}	2599781261562279	c_{33}	14722519233845227915335344	c_{51}	−11228174964373034514526126784	c_{69}
c_{16}	703737238969351	c_{34}	26100016339490264593135930	c_{52}	31947787660331753005976517728	c_{70}
c_{17}	−74689355357780259	c_{35}	−85911886883585776474149659	c_{53}	4525159837076488065598251776	
c_{18}	96224930618884890	c_{36}	−71381602952343952813346574	c_{54}	−24213200715632083926698812032	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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